



MOUNTAINS OF THE MOON UNIVERSITY
FACULTY OF SCIENCE TECHNOLOGY & INOVATIONS
BCS & BIT

Course outline
2023-2024

Course unit	Fundamentals of Structured Programming	
Course Code	BCS 1202	
Credit Units	4	
Lecturer: - Title - E-mail address - Tel	Mr Baranga Peter barangaster@gmail.com 0701-310089	
Course Description	The course gives students the fundamentals of programming. Students learn programme flow, control as well as selected procedures using pseudocodes. Flow charts are used to conceptualize the flow and Syntax of the operations.	
Course Objective (the goal)	i. Give students a conceptual view of the art of programming and programme controls so that they are able to implement programming projects in specific languages. ii. Gain knowledge about the various programming languages iii. Acquire skills in programming concepts iv. Have knowledge in planning and organization of programming projects v. Use techniques of evaluating syntactic and semantic correctness of a computer program Expected Learning Outcomes At the end of this course students will: i. Apply knowledge and skills in programming concepts ii. Plan and organize programming projects iii. Evaluate syntactic and semantic correctness of a computer program iv. Design and implement programs in the programming language that they would have learnt.	
Instruction Method	Lectures, Case studies, Practicals, Peer discussions and Group work	
Assessment	• Individual Assignment/Presentation 10 marks • Group Assignment 10 marks • Central Test 20 marks • Final Exam 60 marks	



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Teaching Hours for Topics

Topic	Name of Topic	Lecture(Hrs)
1.	Introduction to programming languages	3
2.	Program structure	3
3.	Working with Variables and Data types	4
4.	Working with operators & operands	4
5.	Tutorial: Group Presentation (Header Files)	2
6.	Introduction to Functions	4
7.	Program Control statements	6
8.	Tutorial: Group Presentation (Conditional statements)	2
9.	Looping statements	6
10.	Arrays and strings	4
11.	Working with pointers	12
12.	Practical sessions	10
Total Hours		60

Course Outline

Lecture 1	
Date	
Time	
Topic	Introduction to Programming Languages ▪ Introduction to Programming ▪ Requirements ▪ Brief history of C programming language ▪ Steps of preparing a program ▪ The Program development life cycle



	▪ Creating the source code ▪ Compiling the source code ▪ Creating an executable file ▪ Completing the development cycle ▪ Creating our first C program
Readings	
Lecture 2 & 3	
Date	
Time	
Topic	Program Structure ▪ The Main() function ▪ The #include Directive ▪ Variable Definition ▪ Function Prototype ▪ Program statements ▪ Printf() ▪ Scanf() ▪ return ▪ Function Definition ▪ Program comments ▪ Using Braces
Readings	
Lecture 4 & 5	
Date	
Time	
Topic	Working with Variables and Data types ▪ Understanding how computer memory works ▪ Working with Variables ▪ Variable Names ▪ Numeric Variable Types ▪ Variable declaration ▪ Using Typedef ▪ Initializing Numeric Variables ▪ Working with Constants ▪ Literal Constants ▪ Symbolic constants
Readings	
Lecture 6	
Date	
Time	
Topic	Working with Operators and Operands ▪ Statements ▪ Statements and white spaces ▪ Null statements ▪ Compound statements ▪ Expressions



- Simple Expressions
- Complex Expressions
- Operators
- Assignment operator
- Mathematical operators
- Operator precedence and parentheses
- Order of subexpression evaluation
- Relational operators
- The if statement
- The else clause
- Evaluating relational expressions
- Logical operators
- True/False values
- The Precedence of operators
- Compound assignment operator
- Conditional operator
- Comma operator
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Test One

Lecture 7 & 8

Date

Time

Topic

Working with Functions

- What a function is
- How a function works
- Using functions in structured programming
- Writing a function
- Passing arguments to a function
- Function calling

Lecture 9

Date

Time

Topic

Program Control Statements

- Controlling Program Execution
- The for Statement
- Nesting for Statements
- The while Statement
- Nesting while Statements
- The do...while Loop

Readings

Lecture 10 & 11

Date

Time

Topic

Arrays and Strings

- What Is an Array?
- Single-Dimensional Arrays
- Multidimensional Arrays
- Naming and Declaring Arrays
- Initializing Arrays



	▪ Initializing Multidimensional Arrays ▪ Maximum Array Size
Reading	
Lecture 12	
Date	
Time	
Topic	Working with Pointers ▪ What Is a Pointer? ▪ Your Computer's Memory ▪ Creating a Pointer ▪ Pointers and Simple Variables ▪ Declaring Pointers ▪ Initializing Pointers ▪ Using Pointers ▪ Pointers and Variable Types ▪ Pointers and Arrays ▪ The Array Name as a Pointer ▪ Array Element Storage ▪ Pointer Arithmetic ▪ Pointer Caution
	Test Two

Literature:

- i. Brian W., Kernighan (1989). *C Programming Language*, 2nd Edition.
- ii. Samuel P. (2004). *C: A Reference Manual*, 5th Edition
- iii. Robert L.(2010). *Object Oriented Programming in C++*, Third Edition
- iv. Arjun S. (2013). *C++ Programming with Object Oriented Approach*, 1st Edition
- v. Deitel & Deitel, (2007). *C++ How to Program*, 5th Edition